

Types of Cache Mapping

1. Direct Mapping
2. Associative Mapping
3. Set Associative Mapping

1. Direct Mapping

- The direct mapping technique is simple and inexpensive to implement.
- When the CPU wants to access data from memory, it places a address. The index field of CPU address is used to access address.
- The tag field of CPU address is compared with the associated tag in the word read from the cache.
- If the tag-bits of CPU address is matched with the tag-bits of cache, then there is a *hit* and the required data word is read from cache.
- If there is no match, then there is a *miss* and the required data word is stored in main memory. It is then transferred from main memory to cache memory with the new tag.

DIRECT MAPPING

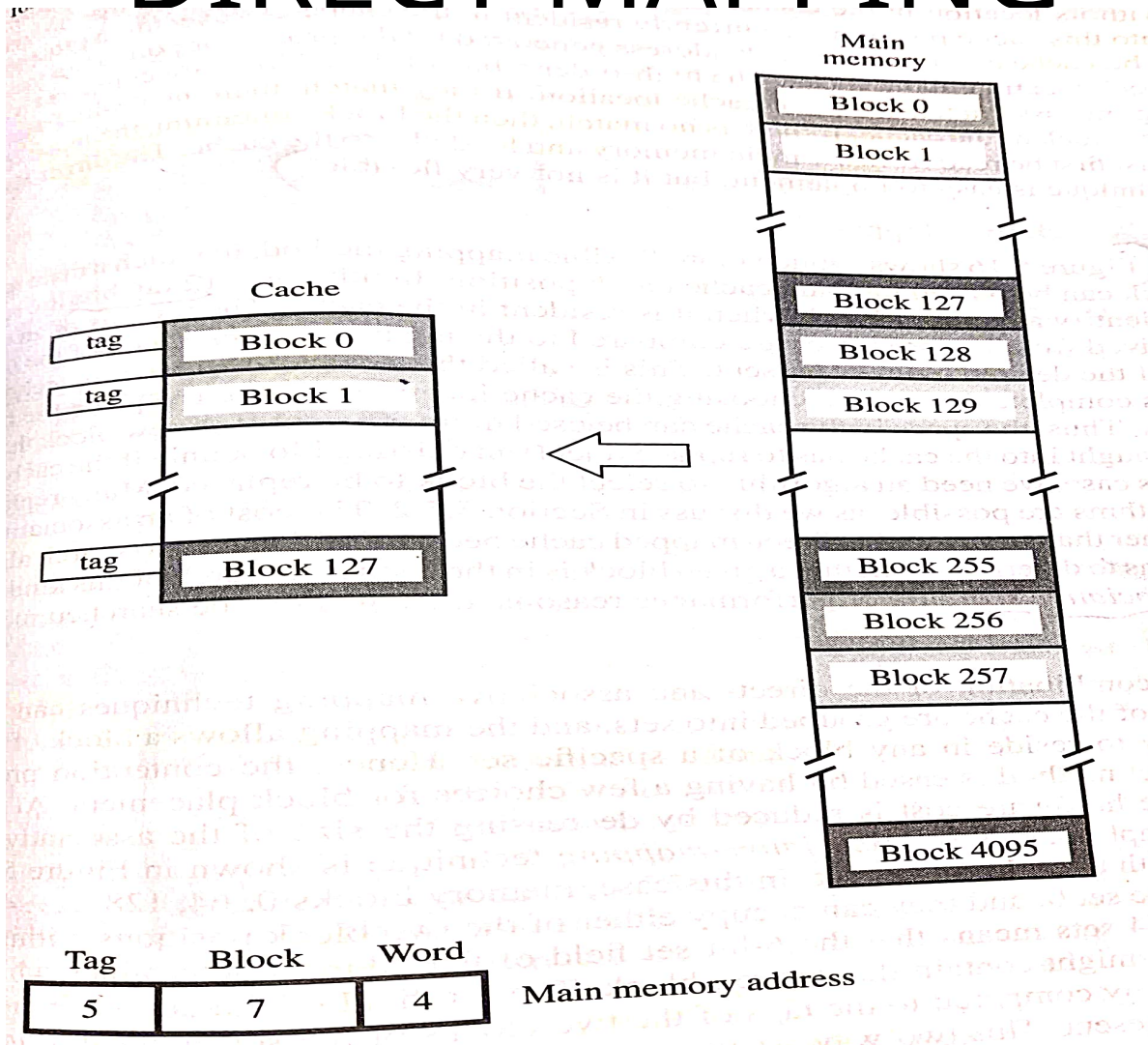


Figure 5.15 Direct-mapped cache.

2. Associative Mapping



- An associative mapping uses an associative memory.
- This memory is being accessed using its contents.
- Each line of cache memory will accommodate the address (main memory) and the contents of that address from the main memory.
- That is why this memory is also called Content Addressable Memory (CAM). It allows each block of main memory to be stored in the cache.

ASSOCIATIVE MAPPING

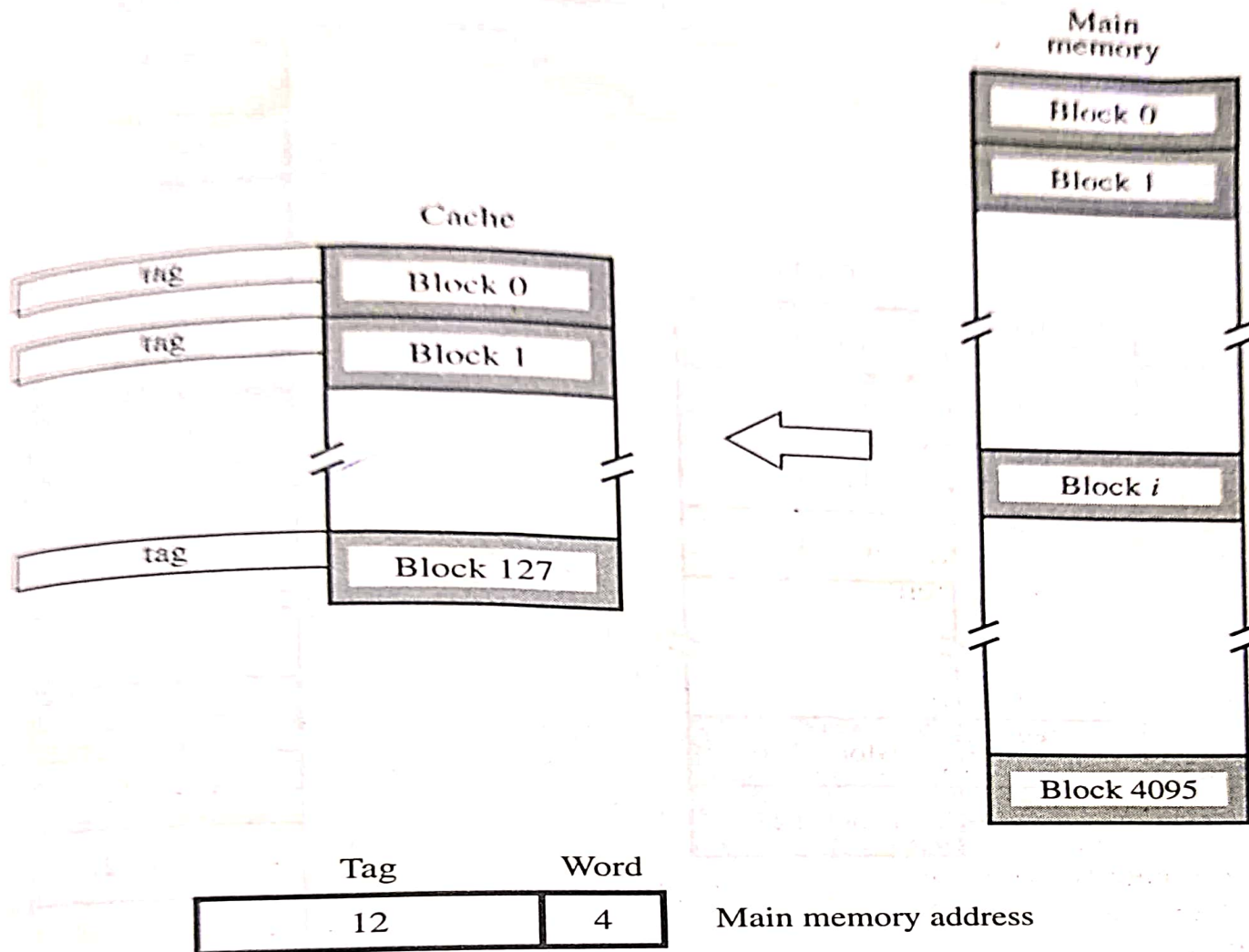
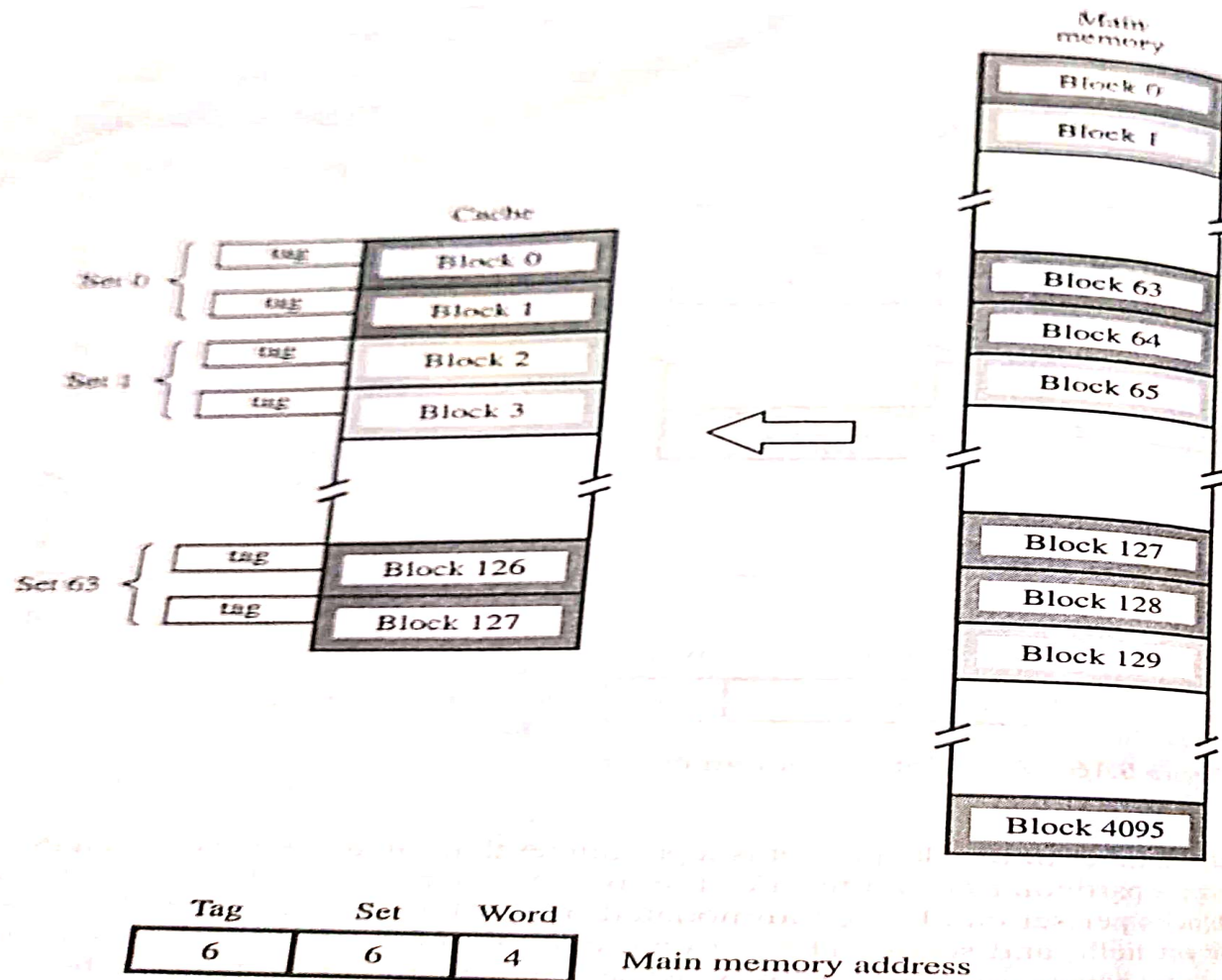


Figure 5.16 Associative-mapped cache.

3. Set Associative Mapping

- That is the easy control of the direct mapping cache and the more flexible mapping of the fully associative cache.
- In set associative mapping, each cache location can have more than one pair of tag + data items.
- That is more than one pair of tag and data are residing at the same location of cache memory. If one cache location is holding two pair of tag + data items, that is called *2-way set associative mapping*.

SET ASSOCIATIVE MAPPING



re 5.17 Set-associative-mapped cache with two blocks per set.

Replacement Algorithms of Cache Memory

- Replacement algorithms are used when there are no available space in a cache in which to place a data. Four of the most common cache replacement algorithms are described below:
 - ***Least Recently Used (LRU):***
 - The LRU algorithm selects for replacement the item that has been least recently used by the CPU.
 - ***First-In-First-Out (FIFO):***
 - The FIFO algorithm selects for replacement the item that has been in the cache from the longest time.
 - ***Random:***
 - The random algorithm selects for replacement the item randomly.
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Writing into Cache

- When memory write operations are performed, CPU first writes into the cache memory. These modifications made by CPU during a write operations, on the data saved in cache, need to be written back to main memory or to auxiliary memory.
- These two popular cache write policies (schemes) are:
 - *Write-Through*
 - *Write-Back*

Write-Through

- In a write through cache, the main memory is updated each time the CPU writes into cache.
- The advantage of the write-through cache is that the main memory always contains the same data as the cache contains.
- This characteristic is desirable in a system which uses direct memory access scheme of data transfer. The I/O devices communicating through DMA receive the most recent data.

Write-Back

- In a write back scheme, only the cache memory is updated during a write operation.
- The updated locations in the cache memory are marked by a flag so that later on, when the word is removed from the cache, it is copied into the main memory.
- The words are removed from the cache time to time to make room for a new block of words.